Denis Bakin

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About me

Computer vision engineer focused on semantic segmentation and weakly-supervised learning. 2 years in a robotics lab (stereo calibration, real-time inference, ROS2/C++); bachelor's thesis on WSSS for plant disease localization at IITP RAS.

Skills

ML/CV: Semantic segmentation, weakly-supervised learning, foundation models (CLIP, SAM), object detection, stereo vision, camera calibration, inference optimization (TorchScript, ONNX)

Technologies: Python, C++, PyTorch / Lightning, Hydra, MLflow, DVC, OpenCV, ROS2, Docker, Git, Linux, GitHub Actions

Languages: Russian (native), English (C1)

Education

Higher School of Economics, Applied Mathematics and Informatics — Sept 2022 – Present

- Specialization: Machine Learning and Applications

Professional Experience

IITP RAS, Color Research Lab, Research Intern — Sept 2025 – Present

- Bachelor's thesis (PDF): "Weakly Supervised Semantic Segmentation for Plant Disease Localization" on PlantSeg (115 disease classes, ~ 7.7k images) + PlantVillage weak labels
- Reproduced MCTformer-V2, PSA/AffinityNet + Random Walk, WeakCLIP; re-evaluated SPDNet (Siamese ADPL-CAM); binary-vs-multiclass framing roughly doubled disease IoU at every pipeline stage
- Fully-supervised SegNeXt upper bound 70.1% disease IoU; proposed a shallow-probe diagnostic exposing $\sim 62\%$ disease IoU latent in SPDNet features (vs. $\sim 32\%$ through the trained classifier); best WSSS+CRF result 38.32% via pre-classifier feature aggregation

HSE Robotics Laboratory, Research Intern — July 2023 – May 2025

- Developed a stereo calibration pipeline: computing intrinsics and extrinsics for a camera pair
- Implemented a frame capture and synchronization pipeline for high-speed cameras via ROS2 (C++)
- Built a detection and segmentation inference pipeline at $100+$ FPS (TorchScript); set up CI with GitHub Actions for lab projects

Projects

Handy — real-time table tennis ball tracking — GitHub

- Team project at HSE Robotics Laboratory. Technologies: Python, C++, OpenCV, PyTorch, ROS2
- Developed a calibration and detection visualization tool (official and full demo)

TouchAndGo — navigation for visually impaired via stereo vision — GitHub

- Depth map generation from a stereo camera pair to simulate 2D lidar and control vibration motors
- Technologies: Python, OpenCV, Raspberry Pi

Other

HSE Lyceum — C++ and Algorithms Teacher; curricula, problem sets, entrance exams — 2024 – Present

Private Tutor — mathematics, computer science, programming — 2022 – Present